Soft drinks/carbonated beverages: HMTc Standard

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CORRESPONDENCE

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for soft drinks/carbonated beverages in Beverages, evaluated on ready-to-drink liquid, as consumed (wet weight, single strength); a concentrate or a powder-to-liquid product is assessed as reconstituted to single strength per the manufacturer's own directions, which is the basis EU 2023/915 and Codex CXS 193 write into their juice cells and Hong Kong Cap 132V Note 11 writes into its beverage entries. 4 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum level for this exact beverage form and analyte in the row's declared native basis, with disclosed read-across where the exact cell is unregulated, a substantiated not-material determination where a species is bounded or is a method artifact, and a control where no defensible sovereign number and no supportable near-neighbour exist. Never occurrence-derived. Ready-to-drink and dry-commodity bases are never mixed and no brewing, dilution or reconstitution factor is invented.

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	200	human-grade read-across (Hong Kong)

Cadmium (Cd)	Tier 1	control	measured every lot; no human-food maximum
Total arsenic (tAs)	Tier 2	control	measured every lot; no human-food maximum
Inorganic arsenic (iAs)	Tier 1	20	human-grade read-across (European Union)
Mercury, total (tHg)	Tier 1	control	measured every lot; no human-food maximum
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	250	human-grade read-across (European Union)
Tin (Sn)	Tier 2	100,000	government maximum level
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	n/m	not material in this matrix

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	200	Hong Kong — human-food analog

Cadmium (Cd)	control	No credible sovereign sets a cadmium maximum level for a soft drink or carbonated beverage. No credible sovereign sets a cadmium maximum level for any formulated non-alcoholic beverage; the EU 2023/915 cadmium cells are commodity cells and the only beverage cadmium values in existence are the packaged--drinking-water standards, which govern water sold as water. No near-neighbour read-across is supportable, so the fail-safe disposition is a control: cadmium testing is mandatory on every certified lot and no ceiling is invented from occurrence data.
Total arsenic (tAs)	control	No credible sovereign sets a total-arsenic maximum level for a soft drink or carbonated beverage. Regulators write beverage arsenic limits on the inorganic species (EU 2023/915 section 3.4.4 fruit juices at 20 ug/kg and section 3.4.1.6 non-alcoholic rice-based drinks at 30 ug/kg; the US FDA apple-juice action level at 10 ug/kg) and write unspeciated arsenic limits only in the packaged-water and public-water-supply schedules, which are Category 8 instruments. The inorganic-arsenic cell in this row carries the health-relevant fraction. Total arsenic is therefore mandatory on every lot as the parent screen and no total-arsenic ceiling is invented.
Inorganic arsenic (iAs)	20	European Union — human-food analog

Mercury, total (tHg)	control	No credible sovereign sets a total-mercury maximum level for a soft drink or carbonated beverage. The sovereign mercury values in this basis family are the packaged-drinking-water quality standards (US 21 CFR 165.110 total mercury at 2 ug/kg; Hong Kong Cap 132V packaged drinking water, written for inorganic mercury at 6 ug/kg) and the fish and seafood methylmercury and total-mercury ceilings, none of which governs this product form. Mercury has no identified enrichment pathway in this supply chain, but the analyte is on the mandatory eight-total panel, so testing is mandatory and no ceiling is invented.
Methylmercury (MeHg)	n/m	Methylmercury has no formation pathway in a soft drink or carbonated beverage: there is no marine bioaccumulation step in the supply chain and no anaerobic, sulfate-reducing methylation environment in the product or its process. Methylmercury is a component of total mercury (MeHg <= tHg) and is governed by and bounded by the total-mercury cell; no separate methylmercury ceiling is published.
Nickel (Ni)	250	European Union — human-food analog
Tin (Sn)	100,000	government maximum level

Aluminium (Al)	control	No credible sovereign sets an aluminium maximum level for a soft drink or carbonated beverage. The only sovereign aluminium values in a dietary basis are the packaged--drinking-water quality standards (US 21 CFR 165.110 secondary allowable level 200 ug/kg, EU 2020/2184, Health Canada 2900 ug/kg), which govern water sold as water and are Category 8 instruments, and the FSANZ infant-formula aluminium limits, which are Category 1 and are excluded from this category's anchor set by the taxonomy boundary rule. In this form aluminium is a packaging and processing signal (aluminium cans and can linings, closures, filter aids, additive carry-over) rather than a raw-material contaminant. Aluminium is a live cell for this row because the dominant packaging format is the aluminium can and because a low-pH carbonated formulation is exactly the condition under which a compromised can lining leaches, so the tripwire is a packaging-integrity control as well as an ingredient control.
Total chromium (Cr)	control	No credible sovereign sets a total-chromium maximum level for a soft drink or carbonated beverage. The only sovereign chromium values reachable from this basis family are the packaged-drinking-water quality standards (US 21 CFR 165.110 at 100 ug/kg, Hong Kong Cap 132V packaged drinking water at 50 ug/kg), which govern water sold as water and are Category 8 instruments, and the public-water-system maximum contaminant levels (US EPA, Health Canada), which are water-utility instruments and not food law at all. Total chromium in this form is dominated by benign Cr(III) and is reported as a stainless-steel equipment and process-water leaching screen, not as a health limit; the health-relevant species is addressed in the Cr-VI cell.

Hexavalent chromium (Cr-VI)	n/m	Hexavalent chromium is not stable in a carbonated or still soft drink. At the pH of this product form (typically pH 2.4 to 3.5, with phosphoric, citric and carbonic acid and invert or high-fructose sugars present in the formulation) Cr(VI) is reduced to Cr(III) by the organic acids, sugars, ascorbate and polyphenols present in the matrix, so a positive Cr(VI) result is an artifact of the total-chromium determination rather than a species present in the beverage as consumed. The reference-method basis for this determination is the in-house species arbitration at data/workbench/standards/-2026-07-31-chromium-formula--adjudication.md, which found that the superseded ion-exchange plus electrothermal-AAS methods that generated the historical Cr(VI) values in liquid food matrices are not reproduced by the modern reference method: saraiva2021 (species-specific isotope dilution HPLC-ICP-MS, limit of quantification 0.049 ug/kg) quantified no Cr(VI) in any milk or formula sample while resolving Cr(III) and total chromium at single-digit ug/kg. That adjudication was drawn on a near-neutral aqueous food matrix; the acidic beverage matrices in this row are a more strongly reducing environment still, so the artifact conclusion carries a fortiori. The health-relevant species is controlled through the total-chromium cell and the reflex rule below, and this determination is a method finding, not an occurrence-derived value: no number is taken from it.

4 | Cite this standard

Pendergrass, K. "Soft drinks/carbonated beverages." In Heavy Metal Tested & Certified (HMTc) Beverages standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

References

[1] European Union — human-food maximum levels
Binds iAs, Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Hong Kong — human-food maximum levels
Binds Pb. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/soft--drinks-carbonated-beverages. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

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